PRANJAL SINGH SESODIA

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EDUCATION

Netaji Subhas University of Technology (NSUT) **Aug 2024 - 2028**
B.Tech. Mechanical Engineering | New Delhi, India ***CGPA 8.07/10 (Dept. Rank: 19/198 - top 10%)***

- **Coursework:** Thermodynamics, Theory of Machines, Solid Mechanics, Control Systems, Fluid Mechanics, Numerical Methods; Ongoing: Robotics, Machine Design

Army Public School Shankar Vihar, New Delhi - CBSE Class XII (PCM): 90%, Class X: 90% **2021 - 2023**

RESEARCH EXPERIENCE

mPRAGATI, Indian Institute of Technology Delhi (IIT Delhi) **Jun 2026 - Present**
R&D Intern | Delhi, India | Supervisor: Dr. Dinesh Kalyanasundaram

- **Designed and fabricated** a **6-DOF robotic arm**; simulated motion in ROS 2 and MuJoCo before hardware testing.
- **Developing encoder-free** closed-loop control for low-cost servos using camera-based joint pose estimation and arm-mounted instrument tracking to drive **PID correction**.
- **Modelled** a patented elbow implant and test jig in **SolidWorks** and analysed ulnar hinge trajectory using ANSYS **multi-body dynamics**; identified a mismatch between simulated and intended motion and **initiated hinge kinematic redesign**.
- **Resized** links of a **polycentric four-bar** prosthetic knee to satisfy the Grashof condition and restore full flexion; verified on a **physical prototype**.

QuEST Lab, IIT Delhi **Dec 2025 - May 2026**
Research Intern | Delhi, India | Supervisor: Dr. Jay Dhariwal

- **Built** a 7-cm-baseline webcam stereo pair for **markerless** surgeon-motion measurement; triangulated distances reached **95-96% accuracy** against measured true distances.
- **Built** surgical-video joint-angle tracking with **AlphaPose** and **limb-line extrapolation** for **occlusion handling**; evaluated Depth Anything monocular depth estimation at **~11 FPS**.
- Co-developed a pump and lining system for saline management in **Minimally Invasive Surgery** (MIS).

PUBLICATION

- **Sesodia, P. S.**, & Dabass, V. (2026). *Robust Real-Time Levitation Gap Estimation for Hyperloop Pods.* Next Research (Elsevier). DOI: 10.1016/j.nexres.2026.102384.

ENGINEERING PROJECTS

Tabletop Subsonic Wind Tunnel **Mar 2026 - Present**
NSUT Delhi | Delhi, India | Supervisor: Dr. Vivek Kumar

- Designed a **6:1 contraction**, optimised it in **ANSYS Fluent** (CFD), used FEA to reduce wall thickness and printed mass, and built drive/measurement electronics.

Catalytic Converter **Nov 2025 - Mar 2026**
Team Daedalus Racing (BAJA SAEINDIA) | NSUT Delhi

- Developed the **team's catalytic converter** using **lower-cost** catalyst materials; analysed **manufacturing process emissions** and sustainability using **Ansys Granta Selector**; contributed to the entry that won **1st nationally** at the mBAJA 2026 Sustainability Event.

Selected Earlier Project
APS Shankar Vihar | India | Mentor: Ms. Tanu Priya Sharma

- **EYEAR:** real-time **speech-to-sign device** (Raspberry Pi, Python); presented to the Union Minister of Education; Indian **provisional patent filed** (2022).

AWARDS & LEADERSHIP

- **1st nationally** and **INR 30,000** award, mBAJA 2026 Sustainability Event; Chattra Pariyojana Vittiya Sahayata (CPVS) project **INR 10,000** grant, NSUT Delhi (2026).
- **Teaching:** instructed **60 students** in a 2-day hands-on 3D-printing workshop, NSUT Innovation and Incubation Foundation (Sept 2026); mentored **8 students** to a Li-ion battery upcycling prototype in 3 weeks, Change.Maker 2.0, Design Dept., IIT Delhi (2025).
- **Daedalus Racing:** founded the Piston Cup Challenge for recruitment; organised events at 3 college fests (250+ participants each).
- **Top 350 nationally**, ATL Marathon; **Smart India Hackathon Junior finalist** (2022).

SKILLS

CAD & Simulation: SolidWorks, ANSYS Fluent (CFD), FEA, ANSYS Multi-Body Dynamics, Ansys Granta Selector, MuJoCo
Robotics & Controls: ROS 2, PID Control, Computer Vision, Pose Estimation, Stereo Camera Calibration, Depth Estimation
Mechanical Design & Fabrication: Linkage Synthesis, Kinematic Analysis, Tolerancing, DFM, FDM/SLA 3D Printing
Programming & Electronics: Python, OpenCV, NumPy, MATLAB, C, Raspberry Pi, ESP32, Arduino